



Chapter 24 - Quality Management

QUALITY CONCEPTS



✧ Software quality & Software standards

Software quality management



- ✧ Concerned with ensuring that the **required level of quality is achieved in a software product.**
- ✧ Three principal concerns:
 - At the **organizational level**, quality management is concerned with **establishing a framework of organizational processes and standards** that will lead to high-quality software.
 - At the **project level**, quality management involves the **application of specific quality processes and checking that these planned processes have been followed.**
 - At the **project level**, quality management is also concerned with **establishing a quality plan for a project.**

Quality management activities



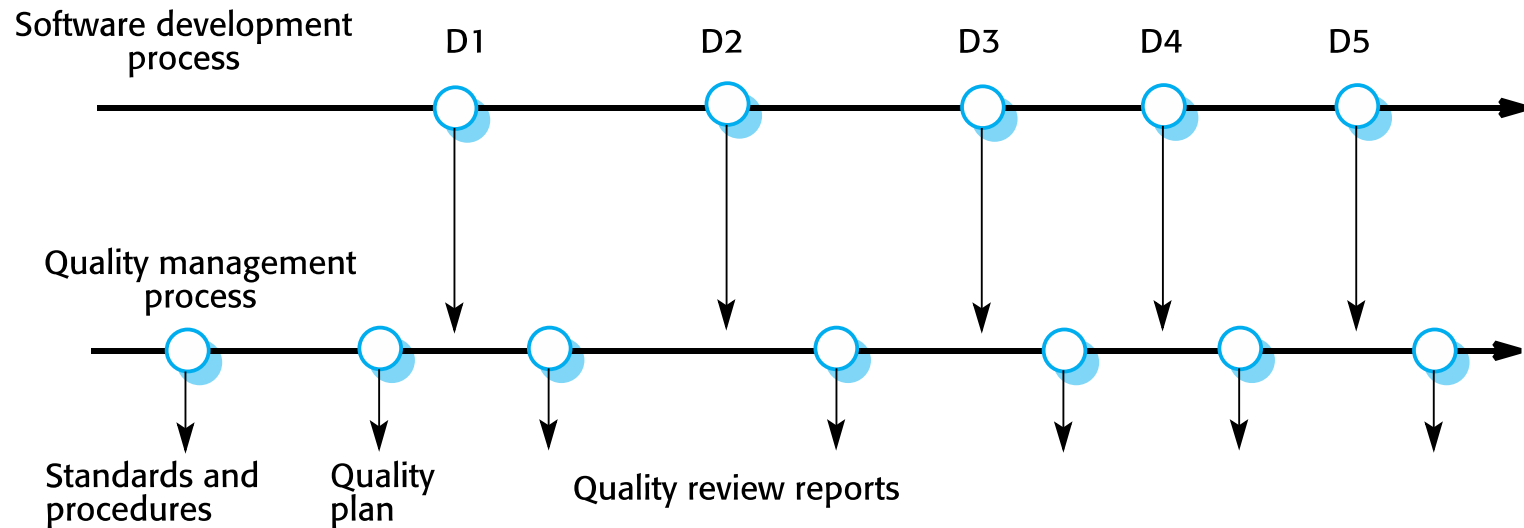
- ✧ Quality management provides an independent check on the software development process.
- ✧ The quality management process checks the project deliverables to ensure that they are consistent with organizational standards and goals
- ✧

Quality management team



- ✧ The **quality team** should be
 - **Independent from the development team** so that they can take an objective view of the software.
- ✧ This allows them to report on software quality without being influenced by software development issues.

Quality management and software development



Quality planning



✧ A quality plan

- sets out the desired product qualities and how these are assessed and defines the most significant quality attributes.
- define the quality assessment process.
- set out which organisational standards should be applied.

Quality plans



✧ Quality plan structure

- Product introduction;
- Product plans;
- Process descriptions;
- Quality goals;
- Risks and risk management.

✧ Quality plans should be short, succinct documents

Scope of quality management



- ✧ Quality management is particularly **important for large, complex systems.**
- ✧ For smaller systems, quality management **needs less documentation**
- ✧ Establishment of “**quality culture**”.

- ✧ **Techniques have to evolve when agile development is used.**



Software quality

Software quality



- ✧ Quality, simplistically, means that a **product should meet its specification**.
- ✧ This is **problematical** for software systems
 - There is a tension between **customer quality** requirements (**efficiency, reliability**, etc.) and **developer quality** requirements (**maintainability, reusability**, etc.);
 - Incomplete & inconsistent software specifications
- ✧ The focus may be '**fitness for purpose**' rather than specification conformance.

Software fitness for purpose



- ✧ Has the software been **properly tested**?
- ✧ Is the software **sufficiently dependable** to be put into use?
- ✧ Is the **performance of the software acceptable** for normal use?
- ✧ Is the **software usable**?
- ✧ Is the software **well-structured and understandable**?
- ✧ Have programming and **documentation standards** been followed in the development process?

Non-functional characteristics



- ✧ The subjective quality of a software system is largely based on its non-functional characteristics.
- ✧ This reflects **practical user experience**.

Software quality attributes



Safety	Understandability	Portability
Security	Testability	Usability
Reliability	Adaptability	Reusability
Resilience	Modularity	Efficiency
Robustness	Complexity	Learnability

Quality conflicts



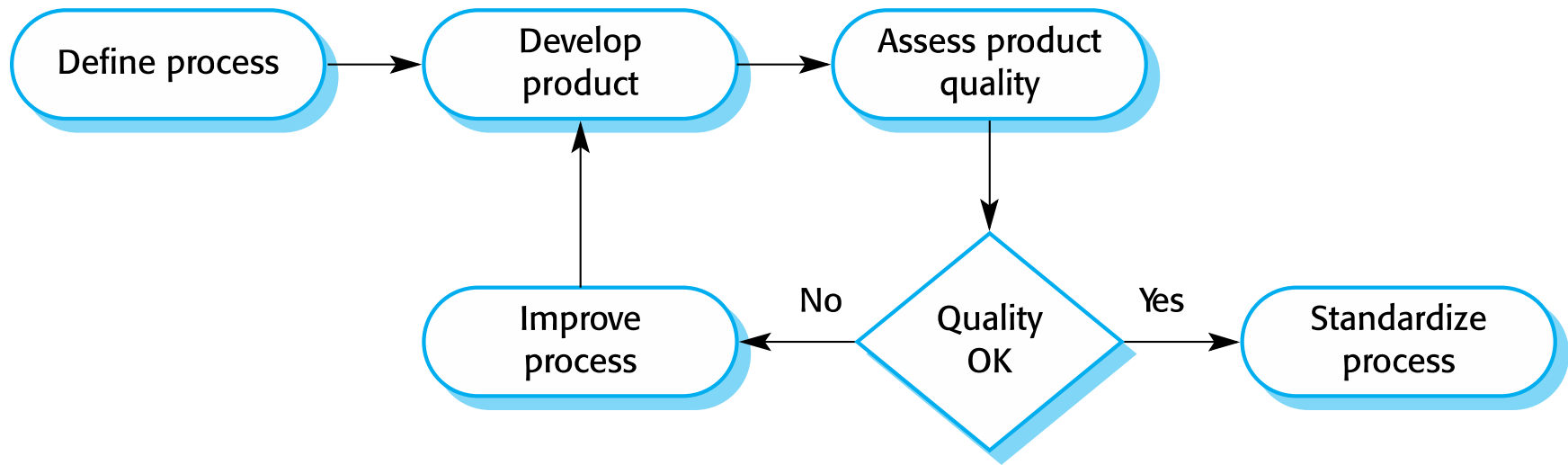
- ✧ Not possible for system to be **optimized for all attributes**
 - for example, **improving robustness** may lead to loss of **performance**.
- ✧ The **quality plan** should
 - define the **most important quality attributes**.
 - include a **definition of the quality assessment process** - maintainability or robustness

Process and product quality



- ✧ However, there is a very complex and poorly understood relationship between **software processes** and **product quality**.
 - The application of **individual skills and experience** is particularly important in **software development**;
 - **External factors** such as the novelty of an application or the need for an accelerated development schedule may **impair product quality**.

Process-based quality



Quality culture



✧ Quality managers should aim to develop a ‘**quality culture**’:

- **everyone responsible** for software development is committed to achieving a high level of product quality.
- encourage **teams to take responsibility** for the quality of their work.
- encourage **professional behavior** in all team members.



Software standards

Software standards



- ✧ Standards define the **required attributes of a product or process**.
 - They play an **important role in quality management**.
- ✧ Standards may be **international, national, organizational or project standards**.

Importance of standards



- ✧ Encapsulation of **best practice** - avoids repetition of past mistakes.
- ✧ They are a **framework for defining what quality means** in a particular setting.
- ✧ They provide **continuity** –
 - new staff can understand the organisation by understanding the standards that are used.

Product and process standards



✧ *Product standards*

- Apply to the **software product** being developed.
- **document standards**, such as the structure of requirements documents
- **coding standards**, which define how a programming language should be used.

✧ *Process standards*

- These define the **processes that should be followed** during software development, which Includes:
 - definitions of specification, design and validation processes.

Product and process standards



Product standards	Process standards
Design review form	Design review conduct
Requirements document structure	Submission of new code for system building
Method header format	Version release process
Java programming style	Project plan approval process
Project plan format	Change control process
Change request form	Test recording process

Problems with standards



- ✧ They may not be seen as relevant and up-to-date by software engineers.
- ✧ They often involve too much bureaucratic form filling.
- ✧ If they are unsupported by software tools, tedious form filling work is often involved to maintain the documentation associated with the standards.

Standards development



✧ Involve practitioners in development.

- Engineers should understand the rationale underlying a standard.

✧ Review standards and their usage regularly.

- Standards can quickly become outdated, and this reduces their credibility amongst practitioners.

✧ Detailed standards should have specialized tool support.

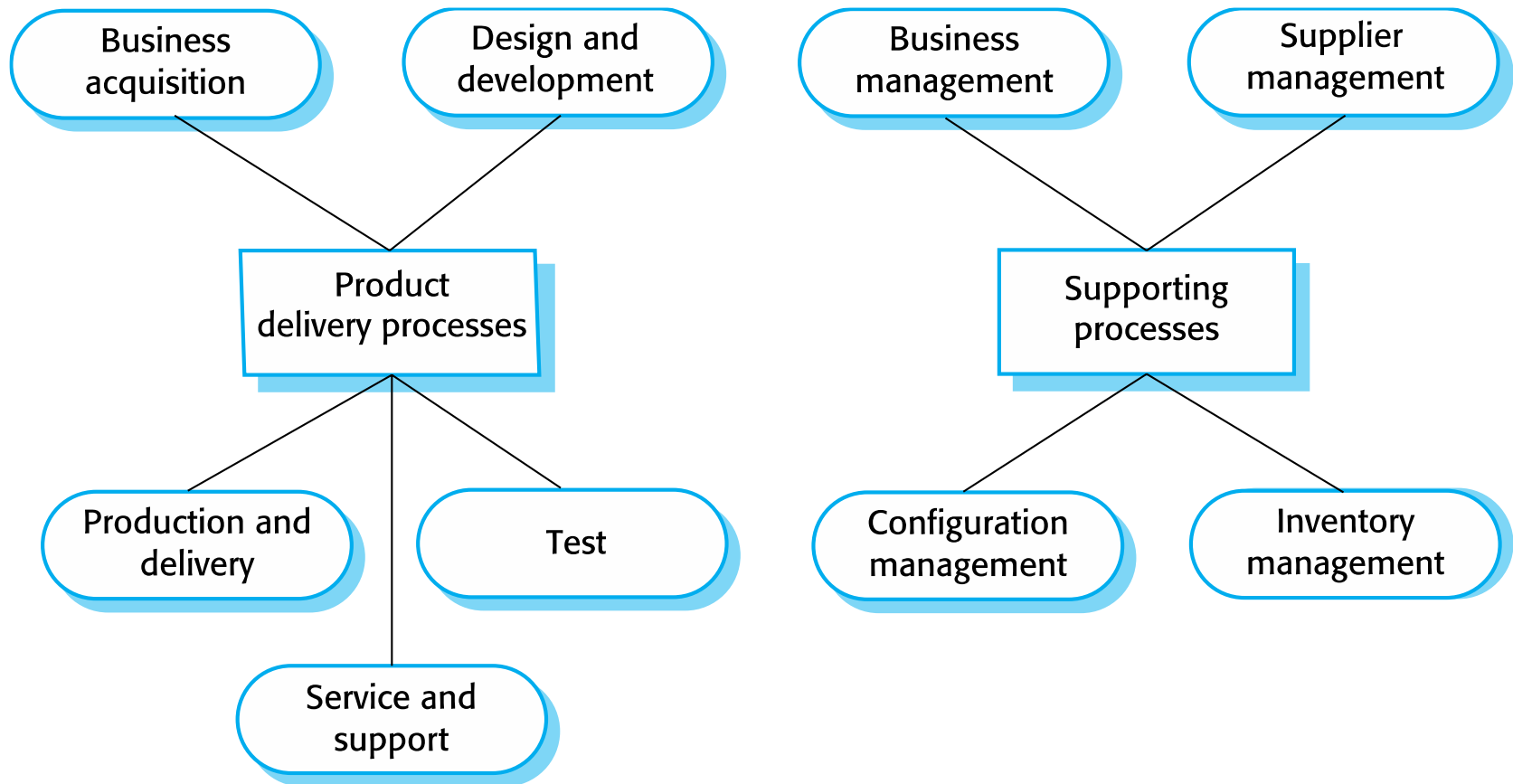
- Excessive clerical work is the most significant complaint against standards.

ISO 9001 standards framework

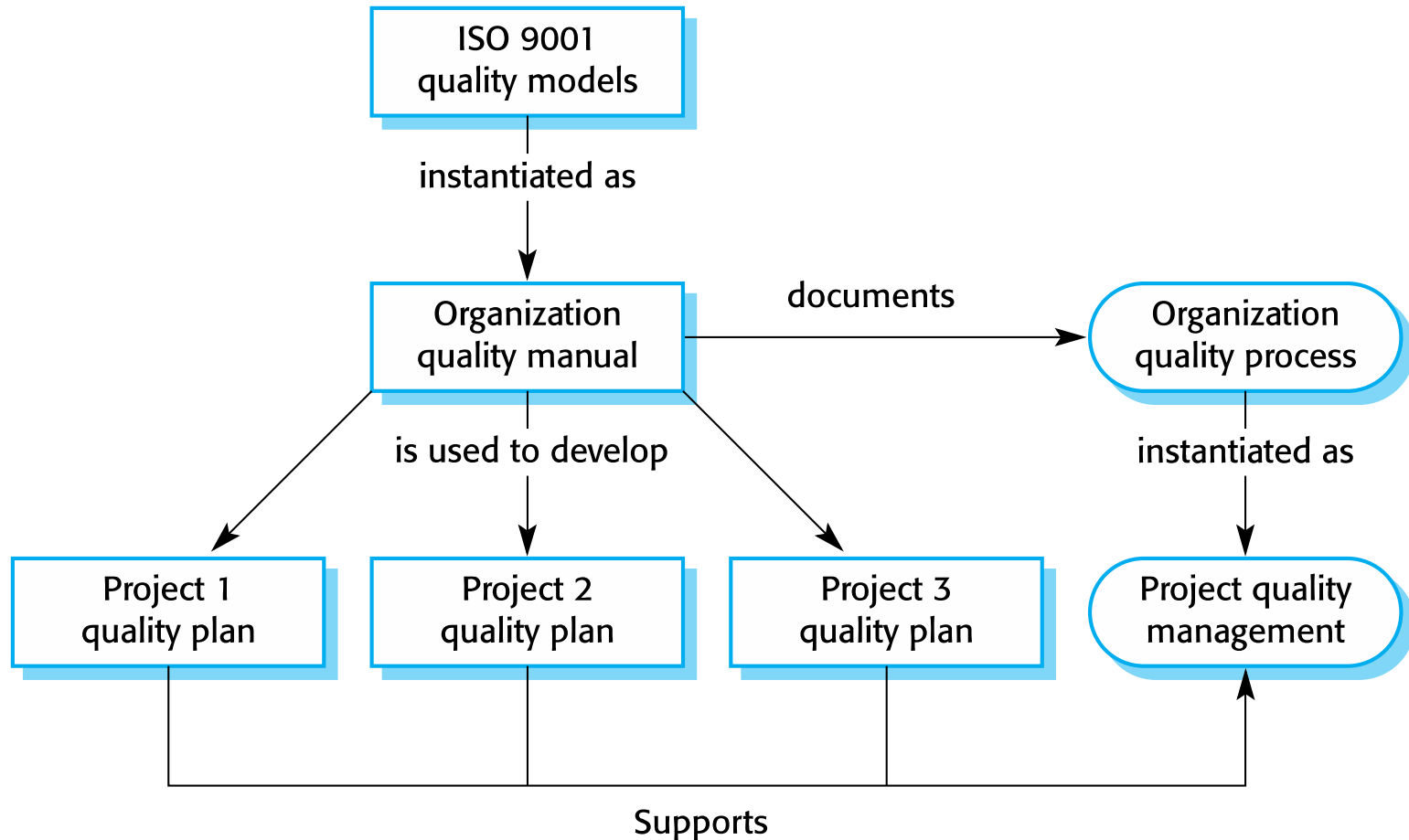


- ✧ An **international set of standards** that can be used as a basis for developing quality management systems.
- ✧ **ISO 9001**, - applies to organizations that design, develop and maintain products, including software.
- ✧ The ISO 9001 standard is a framework for developing software standards.
 - It sets out **general quality principles**, describes **quality processes, etc.**
 - should be documented in an organizational quality manual.

ISO 9001 core processes



ISO 9001 and quality management



ISO 9001 certification



- ✧ Quality standards and procedures should be documented in an organisational quality manual.
- ✧ An external body may certify that an organisation's quality manual conforms to ISO 9000 standards.
- ✧ Some customers require suppliers to be ISO 9000 certified although the need for flexibility here is increasingly recognised.

Software quality and ISO9001



- ✧ The **ISO 9001 certification** is inadequate because:
 - It defines quality to be the conformance to standards.
 - It takes no account of **quality as experienced by users** of the software.
 - standard can be met by incomplete software testing that does not include tests with different method parameters.
- ✧ So long as the defined **testing procedures** are followed and test **records maintained**, the company could be ISO 9001 certified.

Evolution of ISO 9001

